Computer Vision

MCA-574

Lab – 08

BY

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SUBMITTED TO

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Lab 8: Image Enhancement in Medical, Security, Satellite, Photo Restoration, and Biomedical Applications

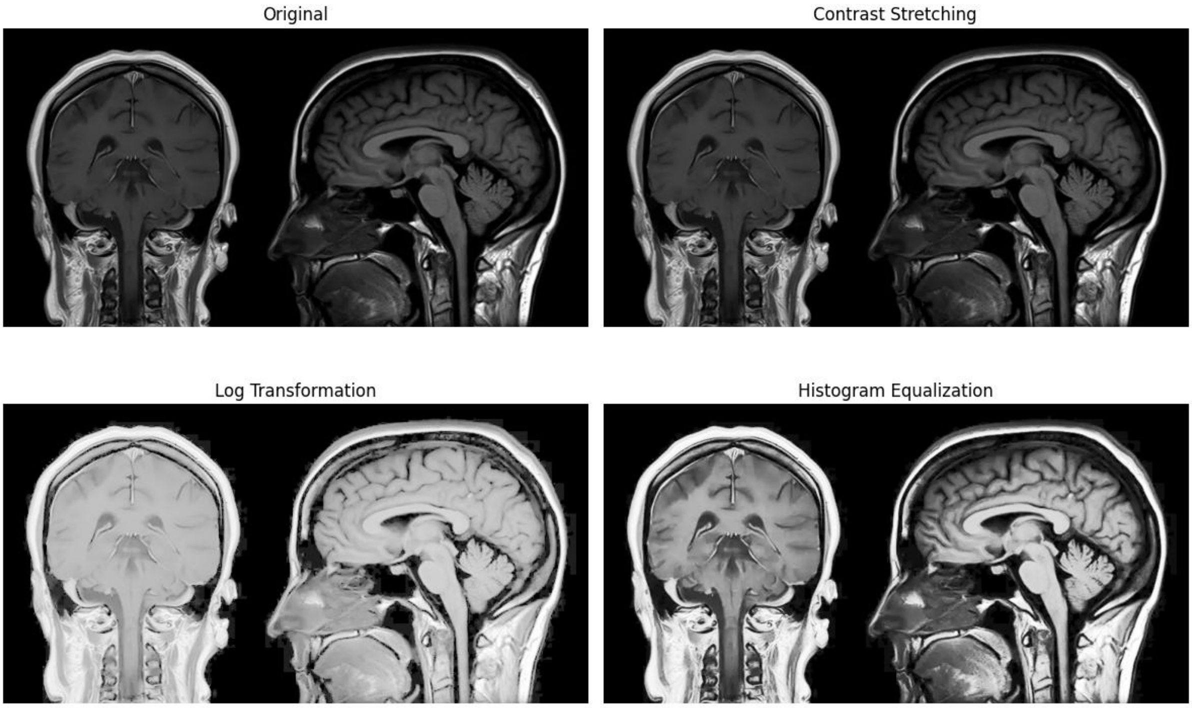
This notebook demonstrates point operations, histogram equalization, and advanced contrast techniques for real-world scenarios.

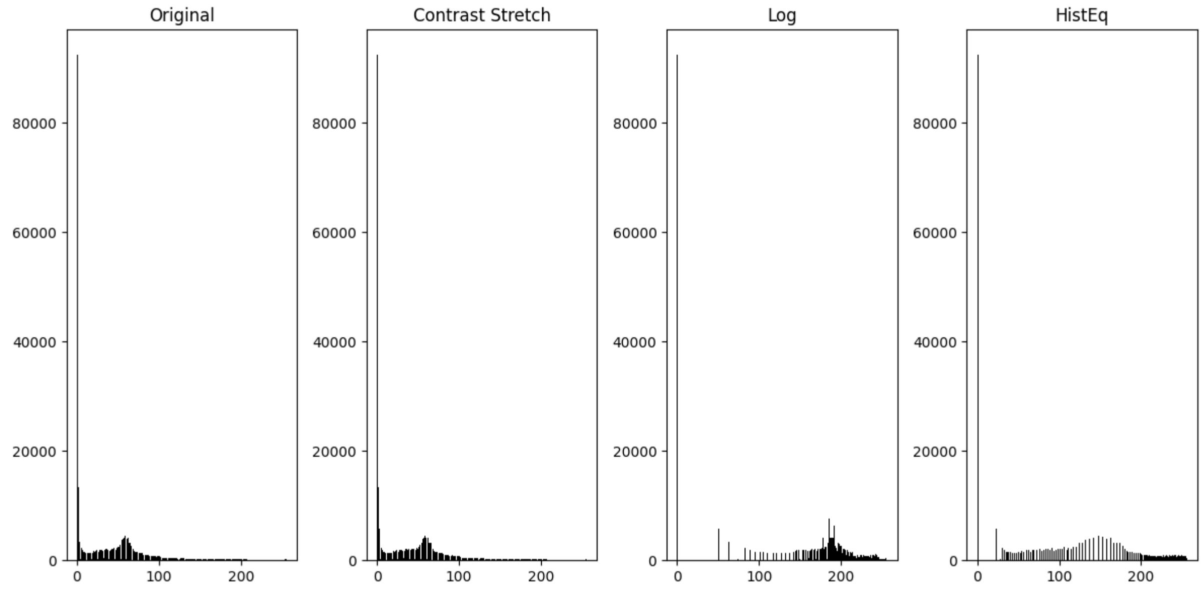
1. Medical Image Enhancement: X-ray Contrast Improvement

Apply contrast stretching, log transformation, and histogram equalization. Plot histograms and write observations.

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load X-ray image (replace with your filename if needed)
img = cv2.imread('mri.jpeg', cv2.IMREAD_GRAYSCALE)
if img is None:
    raise Exception('Image not found!')

# Contrast stretching
min_val, max_val = np.min(img), np.max(img)
cs_img = ((img - min_val) / (max_val - min_val) * 255).astype(np.uint8)

# Log transformation
c = 255 / np.log(1 + np.max(img))
log_img = c * np.log(1 + img.astype(np.float32))
log_img = np.array(log_img, dtype=np.uint8)

# Histogram equalization
hist_eq_img = cv2.equalizeHist(img)

# Plot images
plt.figure(figsize=(12,8))

plt.subplot(2,2,1)
plt.imshow(img, cmap='gray')
plt.title('Original')
plt.axis('off')

plt.subplot(2,2,2)
plt.imshow(cs_img, cmap='gray')
plt.title('Contrast Stretching')
plt.axis('off')

plt.subplot(2,2,3)
plt.imshow(log_img, cmap='gray')
plt.title('Log Transformation')
plt.axis('off')

plt.subplot(2,2,4)
plt.imshow(hist_eq_img, cmap='gray')
```

```
plt.title('Histogram Equalization')
plt.axis('off')
plt.tight_layout()
plt.show()

# Plot histograms
plt.figure(figsize=(12,6))
for i, im in enumerate([img, cs_img, log_img, hist_eq_img]):
    plt.subplot(1,4,i+1)
    plt.hist(im.ravel(), bins=256, range=[0,256], color='black')
    plt.title(['Original','Contrast Stretch','Log','HistEq'][i])
plt.tight_layout()
plt.show()
```





Observations: Medical Image Enhancement

- **Contrast stretching** improves visibility of bone structures by expanding the intensity range, making fractures more distinguishable.
- **Log transformation** enhances low-intensity details, useful for faint features but may compress bright regions.
- **Histogram equalization** provides the most uniform contrast, making overall structures clearer, but may amplify noise in very dark/bright areas.

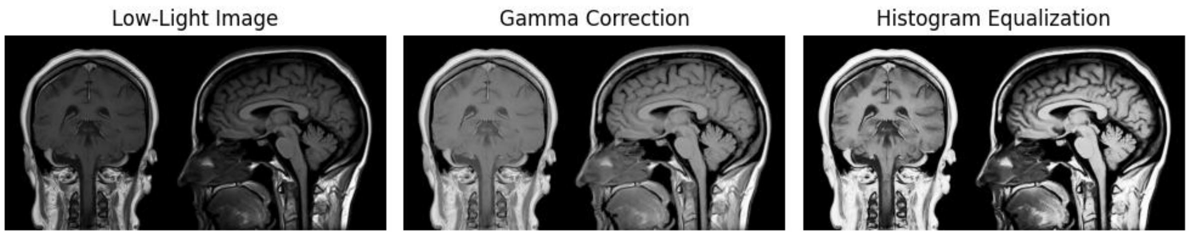
For diagnostic quality, histogram equalization is often preferred, but contrast stretching/log transform can be useful for specific regions.

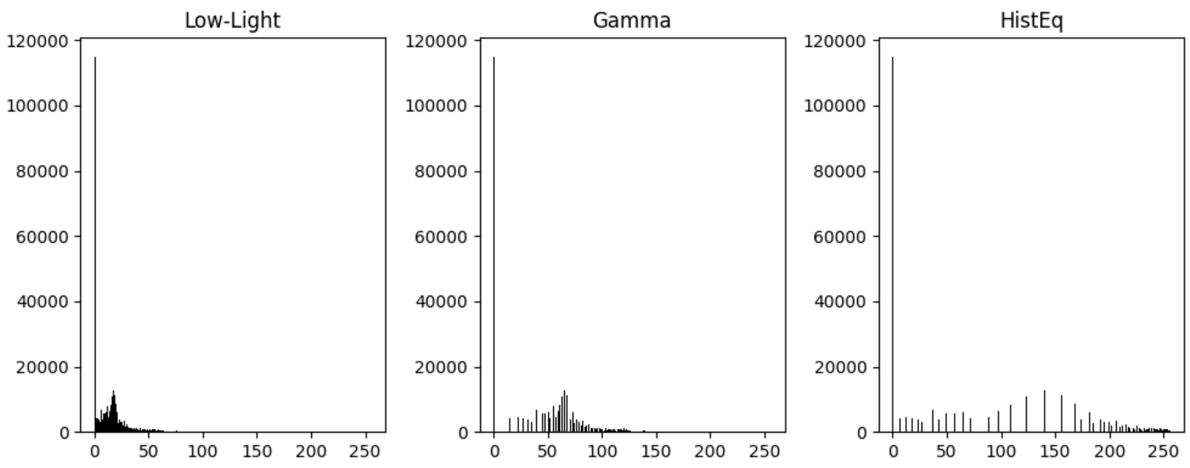
2. Security Surveillance Enhancement: Low-Light Face Visibility

Apply gamma correction and histogram equalization to a low-light image. Compare results and discuss trade-offs.

```
# Simulate a low-light grayscale image (use MRI as example)
img = cv2.imread('mri.jpeg', cv2.IMREAD_GRAYSCALE)
if img is None:
    raise Exception('Image not found!')
low_light = (img * 0.3).astype(np.uint8)

# Gamma correction (brighten dark regions)
gamma = 2.0
gamma_corr = np.array(255 * (low_light / 255) ** (1/gamma), dtype=np.uint8)

# Histogram equalization
hist_eq = cv2.equalizeHist(low_light)
```

```python
# Plot images
plt.figure(figsize=(10,6))
plt.subplot(1,3,1)
plt.imshow(low_light, cmap='gray')
plt.title('Low-Light Image')
plt.axis('off')

plt.subplot(1,3,2)
plt.imshow(gamma_corr, cmap='gray')
plt.title('Gamma Correction')
plt.axis('off')

plt.subplot(1,3,3)
plt.imshow(hist_eq, cmap='gray')
plt.title('Histogram Equalization')
plt.axis('off')
plt.tight_layout()
plt.show()

# Plot histograms
plt.figure(figsize=(10,4))
for i, im in enumerate([low_light, gamma_corr, hist_eq]):
    plt.subplot(1,3,i+1)
    plt.hist(im.ravel(), bins=256, range=[0,256], color='black')
    plt.title(['Low-Light','Gamma','HistEq'][i])
plt.tight_layout()
plt.show()
```





Observations: Security Surveillance Enhancement

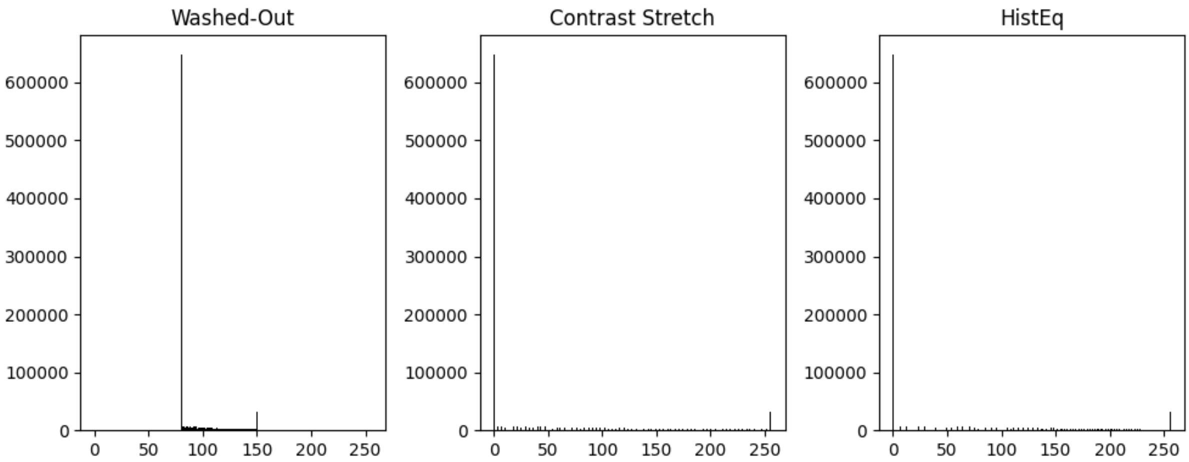
- **Gamma correction** brightens dark regions without over-saturating bright areas, improving facial visibility in low-light images.
- **Histogram equalization** increases overall contrast but may introduce noise or over-brighten some regions.
- For facial identification, gamma correction is often preferred for natural enhancement, while histogram equalization is useful for general contrast improvement.
- **Trade-off:** Gamma preserves details in shadows, histogram equalization may lose subtle features in very dark/bright areas.

3. Satellite Image Contrast Correction

Apply contrast stretching and histogram equalization to a washed-out satellite image. Analyze differentiation between vegetation and barren land.

```
# Load satellite image (replace with your filename if needed)
sat_img = cv2.imread('moon.webp', cv2.IMREAD_GRAYSCALE)
if sat_img is None:
    raise Exception('Image not found!')

# Simulate washed-out effect (intensities 80-150)
sat_img = np.clip(sat_img, 80, 150)

# Contrast stretching
cs_sat = ((sat_img - 80) / (150 - 80) * 255).astype(np.uint8)

# Histogram equalization
hist_eq_sat = cv2.equalizeHist(sat_img)

# Plot images
plt.figure(figsize=(10,6))

plt.subplot(1,3,1)
plt.imshow(sat_img, cmap='gray')
plt.title('Washed-Out Image')
plt.axis('off')

plt.subplot(1,3,2)
plt.imshow(cs_sat, cmap='gray')
plt.title('Contrast Stretching')
plt.axis('off')

plt.subplot(1,3,3)
plt.imshow(hist_eq_sat, cmap='gray')
plt.title('Histogram Equalization')
plt.axis('off')
```

```
plt.tight_layout()
plt.show()

# Plot histograms
plt.figure(figsize=(10,4))
for i, im in enumerate([sat_img, cs_sat, hist_eq_sat]):
    plt.subplot(1,3,i+1)
    plt.hist(im.ravel(), bins=256, range=[0,256], color='black')
    plt.title(['Washed-Out','Contrast Stretch','HistEq'][i])
plt.tight_layout()
plt.show()
```



Observations: Satellite Image Contrast Correction

- **Contrast stretching** redistributes pixel intensities, making differences between vegetation and barren land more visible.
- **Histogram equalization** enhances global contrast, but may exaggerate noise or small features.
- For agricultural analysis, contrast stretching is often preferred for controlled enhancement, while histogram equalization is useful for general contrast improvement.

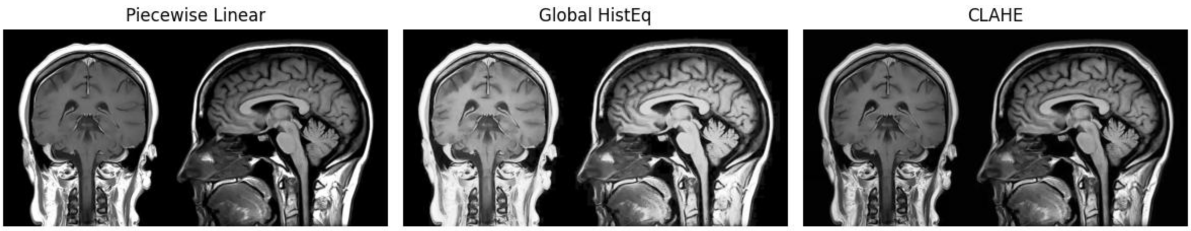
- Visual and histogram analysis show better separation of land types after enhancement.

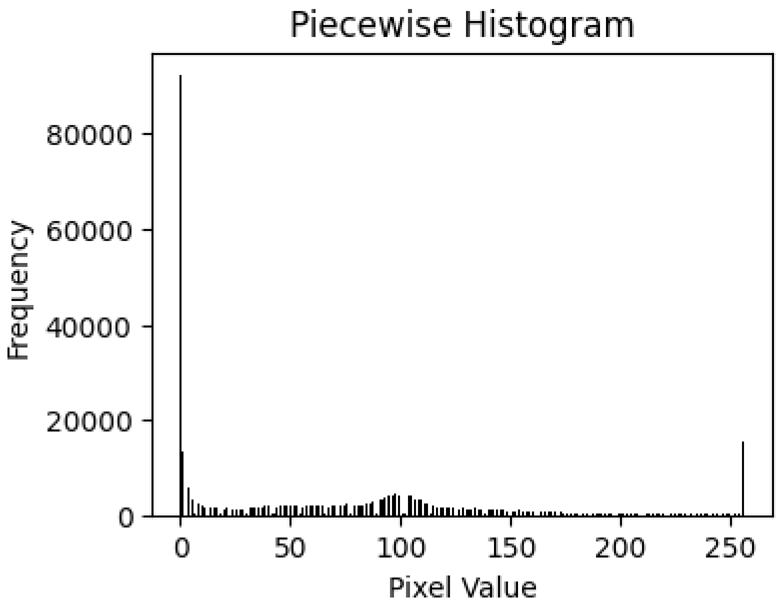
4. Digital Photo Restoration: Archival Enhancement

Apply piecewise linear transformation and adaptive histogram equalization (CLAHE). Compare global histogram equalization vs CLAHE visually and numerically.

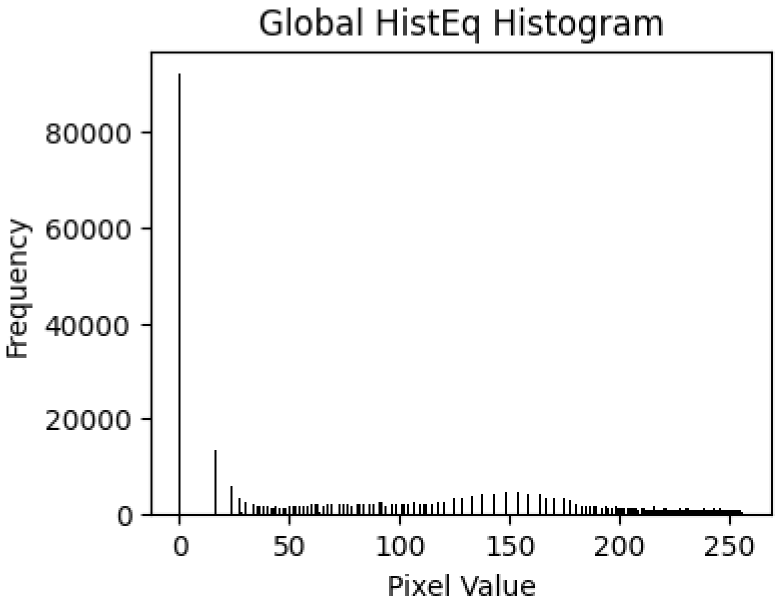
```
# Simulate a scanned photo (use MRI as example)
photo = cv2.imread('mri.jpeg', cv2.IMREAD_GRAYSCALE)
if photo is None:
    raise Exception('Image not found!')

# Piecewise linear transformation: contrast stretching + thresholding
min_val, max_val = np.percentile(photo, [5, 95])
piecewise = np.clip((photo - min_val) / (max_val - min_val) * 255, 0,
255).astype(np.uint8)
piecewise[photo < min_val] = 0
piecewise[photo > max_val] = 255

# Global histogram equalization
glob_hist_eq = cv2.equalizeHist(photo)

# Adaptive histogram equalization (CLAHE)
clahe = cv2.createCLAHE(clipLimit=2.0, tileGridSize=(8,8))
adapt_hist_eq = clahe.apply(photo)

# Plot images
plt.figure(figsize=(12,6))

plt.subplot(1,3,1)
plt.imshow(piecewise, cmap='gray')
plt.title('Piecewise Linear')
plt.axis('off')

plt.subplot(1,3,2)
plt.imshow(glob_hist_eq, cmap='gray')
plt.title('Global HistEq')
plt.axis('off')

plt.subplot(1,3,3)
plt.imshow(adapt_hist_eq, cmap='gray')
plt.title('CLAHE')
plt.axis('off')
plt.tight_layout()
plt.show()

# Compare histograms, mean, std
for name, im in zip(['Piecewise','Global HistEq','CLAHE'], [piecewise,
glob_hist_eq, adapt_hist_eq]):
```

```
    plt.figure(figsize=(4,3))
    plt.hist(im.ravel(), bins=256, range=[0,256], color='black')
    plt.title(f'{name} Histogram')
    plt.xlabel('Pixel Value')
    plt.ylabel('Frequency')
    plt.show()
    print(f'{name}: Mean={np.mean(im):.2f}, Std={np.std(im):.2f}')
```

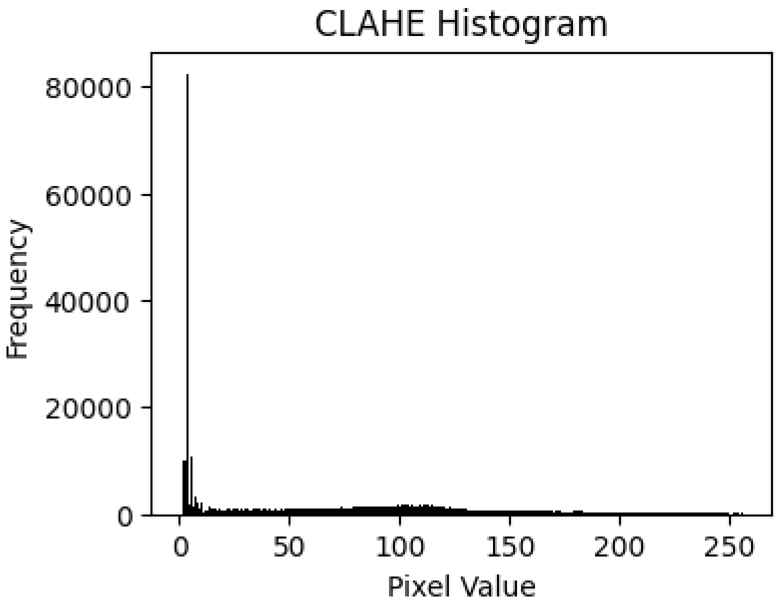




Piecewise: Mean=66.68, Std=73.93



Global HistEq: Mean=89.66, Std=84.90



CLAHE: Mean=65.11, Std=64.77

Observations: Digital Photo Restoration

- **Piecewise linear transformation** restores overall brightness and contrast, but may lose local details.
- **Global histogram equalization** improves contrast globally, but can cause over-enhancement in some regions.

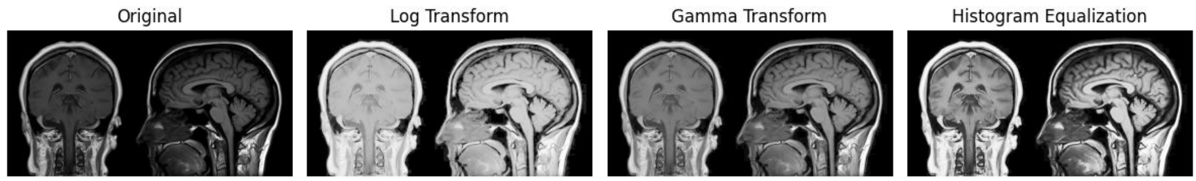
- **CLAHE** (adaptive histogram equalization) enhances local contrast, preserves details, and avoids over-saturation.
- Numerically, CLAHE provides better histogram spread and balanced mean/std, making it most suitable for photo quality preservation.

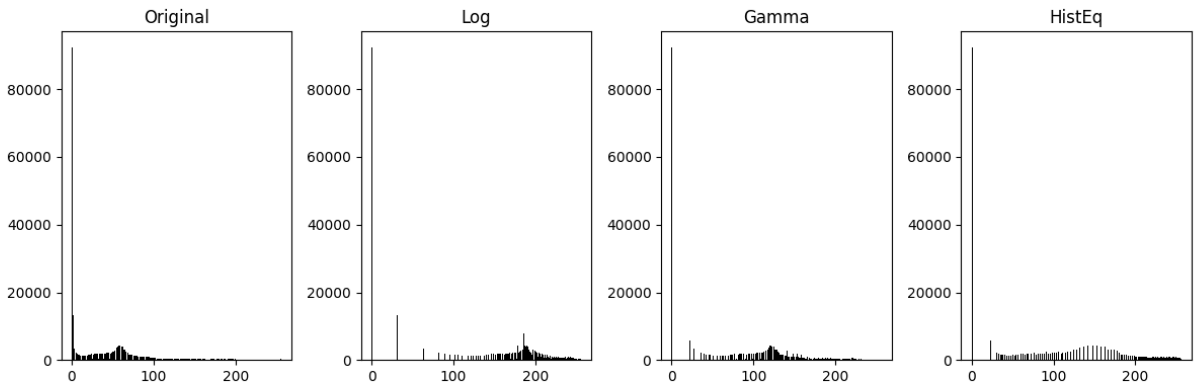
5. Biomedical Microscopic Image Enhancement

Apply log and gamma transformations, then histogram equalization. Compare clarity of cell edges and discuss which method is best for fine detail enhancement.

```
# Simulate a cell image (use MRI as example)
cell_img = cv2.imread('mri.jpeg', cv2.IMREAD_GRAYSCALE)
if cell_img is None:
    raise Exception('Image not found!')

# Log transformation
c = 255 / np.log(1 + np.max(cell_img))
log_cell = c * np.log(1 + cell_img.astype(np.float32))
log_cell = np.array(log_cell, dtype=np.uint8)

# Gamma transformation
gamma = 0.5
gamma_cell = np.array(255 * (cell_img / 255) ** gamma, dtype=np.uint8)

# Histogram equalization
hist_eq_cell = cv2.equalizeHist(cell_img)

# Plot images
plt.figure(figsize=(12,6))

plt.subplot(1,4,1)
plt.imshow(cell_img, cmap='gray')
plt.title('Original')
plt.axis('off')

plt.subplot(1,4,2)
plt.imshow(log_cell, cmap='gray')
plt.title('Log Transform')
plt.axis('off')

plt.subplot(1,4,3)
plt.imshow(gamma_cell, cmap='gray')
plt.title('Gamma Transform')
plt.axis('off')

plt.subplot(1,4,4)
plt.imshow(hist_eq_cell, cmap='gray')
plt.title('Histogram Equalization')
plt.axis('off')
```

```python
plt.tight_layout()
plt.show()

# Plot histograms
plt.figure(figsize=(12,4))
for i, im in enumerate([cell_img, log_cell, gamma_cell, hist_eq_cell]):
    plt.subplot(1,4,i+1)
    plt.hist(im.ravel(), bins=256, range=[0,256], color='black')
    plt.title(['Original','Log','Gamma','HistEq'][i])
plt.tight_layout()
plt.show()
```





Observations: Biomedical Microscopic Image Enhancement

- **Log and gamma transformations** enhance small intensity variations, making cell boundaries more visible.
- **Histogram equalization** improves overall contrast but may not highlight fine details as effectively as log/gamma transforms.
- For fine detail enhancement, log/gamma transformations are preferred; histogram equalization is better for global contrast.
- Histogram plots and image comparison support these conclusions.